

For new file upload: Init → Add → Commit → Push

For modified file upload: Add → Commit → Push

1. change directory

- **Usage:** cd..
- **Explanation:** go back to the directory
- **Usage:** cd <folder name>
- **Explanation:** make new directory folder

2. git init

- **Usage:** git init
- **Explanation:** Initializes a new Git repository in the current directory. It sets up all the necessary files for Git to start tracking changes. (clone local repository with GitHub)

3. git clone

- **Usage:** git clone <repository-url>
- **Explanation:** Clones a repository from a remote source (e.g., GitHub) to your local machine. This downloads all the repository's history and files.

Or

4. git remote

- **Usage:** git remote add <name> <url>
- **Explanation:** Manages remote connections to a repository. You can add, remove, and view remotes like origin.

5. git add

- **Usage:** `git add <file>` or `git add.`
- **Explanation:** Stages changes (adds them to the index) so they can be committed. Use `git add.` to add all changed files in the current directory.

(Git add. → use to move all file in one time to local repository)

6. git commit

- **Usage:** `git commit -m "message"`
- **Explanation:** Records changes to the repository with a message describing what has been done. It saves the current state of your staged changes.

(to send local git folder)

7. git status

- **Usage:** `git status`
- **Explanation:** Displays the current state of the working directory and the staging area. Shows which files have been modified, which are staged for the next commit, and which aren't being tracked.

8. git push

- **Usage:** `git push origin <branch>`
- **Explanation:** Pushes the local commits to the remote repository. You need to specify which branch you're pushing to (e.g., `main` or `master`).

(upload a file from local git folder to GitHub Repository)

9. git pull

- **Usage:** git pull origin <branch>
- **Explanation:** Fetches the latest changes from the remote repository and merges them with your current branch.

(Get all updates from GitHub to Local git folder)

10.git branch

- **Usage:** git branch or git branch <branch-name>
- **Explanation:** Lists all local branches or creates a new branch. Branches allow multiple versions of a project to be worked on simultaneously.

11.git checkout

- **Usage:** git checkout <branch-name> or git checkout <commit-hash>
- **Explanation:** Switches to a different branch or restores files from a specific commit.

12.git merge

- **Usage:** git merge <branch-name>
- **Explanation:** Merges another branch into the current branch. It combines the changes from the specified branch into your current branch.

13.git fetch

- **Usage:** git fetch origin
- **Explanation:** Retrieves changes from the remote repository without merging them into your local branch. This updates your remote tracking branches.

14.git log

- **Usage:** git log
- **Explanation:** Shows a log of commits for the current branch. It displays information such as commit IDs, authors, and messages.

15.git reset

- **Usage:** git reset <file> or git reset --hard <commit>
- **Explanation:** Unstages files from the index or resets the working directory to a previous commit. Using --hard discards all changes after the specified commit.

16.git revert

- **Usage:** git revert <commit>
- **Explanation:** Creates a new commit that undoes the changes made by a previous commit. Unlike reset, it does not alter the commit history.

17.git stash

- **Usage:** git stash or git stash pop
- **Explanation:** Temporarily saves changes in a "stash" without committing them. You can use git stash pop to retrieve them later.

18.git diff

- **Usage:** git diff
- **Explanation:** Displays the differences between your working directory and the index or between different commits.

19.git rm

- **Usage:** git rm <file>
- **Explanation:** Removes files from the working directory and the staging area.

20.git rebase

- **Usage:** git rebase <branch>
- **Explanation:** Re-applies commits on top of another base tip. It moves or combines a sequence of commits to a new base commit.

21.git tag

- **Usage:** git tag <tagname>
- **Explanation:** Tags a specific commit with a name, often used to mark release points (e.g., v1.0).

22.git show

- **Usage:** git show <commit>
- **Explanation:** Displays detailed information about a specific commit, including the changes made, author information, and timestamps.